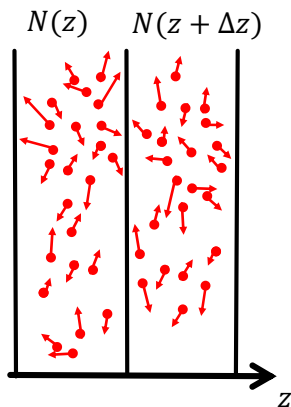


Diffusion approximation

Basic concepts

- ▶ At some point, deep enough in the tissue, we may assume that the radiance $I(\mathbf{r}, \hat{\mathbf{s}})$ is almost isotropic
- ▶ In terms of Monte-Carlo simulations, we may assume that the photon-packets are randomly going in all directions
- ▶ Analogy may be drawn to particles in gases or temperature distribution (diffusive processes)
- ▶ The directional dependence (almost) stops being interesting, and we care more about angle-integrated quantities, e.g. $\int_{(4\pi)} I(\mathbf{r}, \hat{\mathbf{s}}) d\Omega$.

Basics of diffusion – Fick's law



N – number of particles in slab

$u = N/V$ – density of particles

J – flux of particles

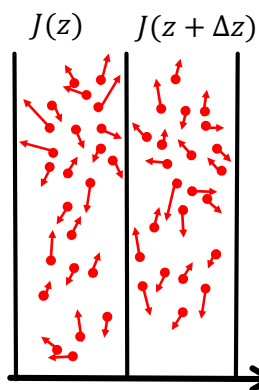
Δt – The time it takes the particles to leave the slab

$$J = -\frac{1}{2}[N(z) - N(z + \Delta z)] \times \frac{1}{A\Delta t}$$

Fick's law: flux goes according to the gradient in density

$$J = -D \frac{d}{dz} u(z)$$

Conservation law (mass, energy, etc.)



$$\Delta N(z,t) = [J(z,t) - J(z + \Delta z,t)]A\Delta t$$

Conservation law:

$$\frac{\partial}{\partial t} u(z,t) = -\frac{\partial}{\partial z} J(z,t)$$

Diffusion equation

- Combining Fick's law with the conservation law, we obtain

$$\frac{\partial}{\partial t} u(z, t) = D \frac{\partial^2}{\partial z^2} u(z, t)$$

- And in 3D:

$$\frac{\partial}{\partial t} u(z, t) = D \nabla^2 u(z, t)$$

Since every physical system has its own conservation, the diffusion equation is a direct result of Fick's law

Conservation law in RTE

RTE:

$$\frac{1}{c} \frac{\partial I(\mathbf{r}, \hat{\mathbf{s}})}{\partial t} + \hat{\mathbf{s}} \cdot [\nabla I(\mathbf{r}, \hat{\mathbf{s}})] + \mu_t I(\mathbf{r}, \hat{\mathbf{s}}) - \mu_t \int_{4\pi} I(\mathbf{r}, \hat{\mathbf{s}}') p(\hat{\mathbf{s}}', \hat{\mathbf{s}}) d\Omega' = \epsilon(r, \hat{\mathbf{s}})$$

Energy density and flux:

$$U(\mathbf{r}) = \int_{(4\pi)} I(\mathbf{r}, \hat{\mathbf{s}}) d\Omega \quad \mathbf{J}(\mathbf{r}) = \int_{(4\pi)} I(\mathbf{r}, \hat{\mathbf{s}}) \hat{\mathbf{s}} d\Omega$$

Integrating the RTE over 4π , we obtain the conservation relation:

$$\frac{1}{c_0} \frac{\partial}{\partial t} U(\mathbf{r}) + \mu_a U(\mathbf{r}) + \nabla \cdot \mathbf{J}(\mathbf{r}) = S_0(\mathbf{r})$$

Decomposition of $I(\mathbf{r}, \hat{\mathbf{s}})$ in 1D

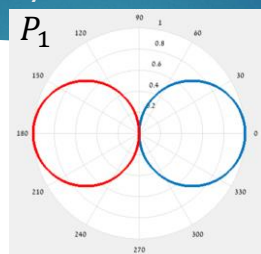
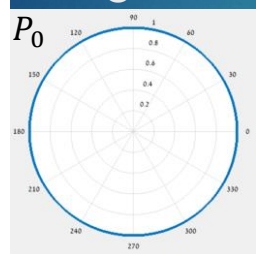
- In the 1D case, we showed that the radiance may be decomposed in $\tau = \cos\theta$ to Legendre polynomials, which are orthogonal

$$\psi(x, \tau) = \sum_{n=0}^{\infty} \frac{2n+1}{2} \phi_n(x) P_n(\tau)$$

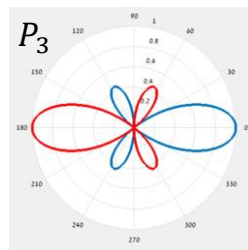
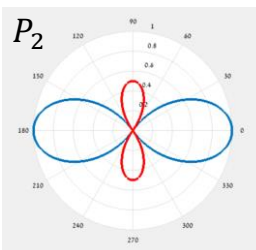
n	$P_n(x)$
0	1
1	x
2	$(3x^2 - 1)$

- If we perform $\int_{-1}^1 d\tau$, we get that the energy density is equal to $\phi_0(x)$, and if we perform $\int_{-1}^1 \tau d\tau$, we get that the flux is equal to with $\phi_1(x)$.
- If we wish to derive a “diffusion like” equation for the flux and energy, only a first order approximation is needed

Visual representation of the Legendre polynomials



Blue is positive
Red is negative



Only $P_0(x)$ has energy
Only $P_1(x)$ has flux

Decomposition of $I(\mathbf{r}, \hat{\mathbf{s}})$ in 3D

- In 3D, the decomposition is to spherical harmonics, but the rationale remains

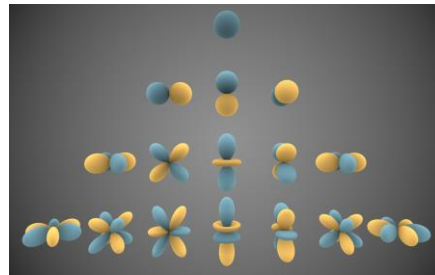
$$I(\mathbf{r}, \hat{\mathbf{s}}) = \sum_{n=0}^{\infty} \sum_{m=-n}^{m=n} \phi_{n,m}(\mathbf{r}) Y_{n,m}(\hat{\mathbf{s}}) \approx \sum_{n=0}^1 \sum_{m=-n}^{m=n} \phi_{n,m}(\mathbf{r}) Y_{n,m}(\hat{\mathbf{s}})$$

$$Y_0^0(\theta, \varphi) = \frac{1}{2} \sqrt{\frac{1}{\pi}}$$

$$Y_1^{-1}(\theta, \varphi) = \frac{1}{2} \sqrt{\frac{3}{2\pi}} \cdot e^{-i\varphi} \cdot \sin \theta$$

$$Y_1^0(\theta, \varphi) = \frac{1}{2} \sqrt{\frac{3}{\pi}} \cdot \cos \theta$$

$$Y_1^1(\theta, \varphi) = \frac{-1}{2} \sqrt{\frac{3}{2\pi}} \cdot e^{i\varphi} \cdot \sin \theta$$



The diffusion approximation

- The diffused radiance includes an “energy term” ($n=0$) and a “flux term” ($n=1$)

$$I_d(\mathbf{r}, \hat{\mathbf{s}}) = \sum_{n=0}^1 \sum_{m=-n}^{m=n} \phi_{n,m}(\mathbf{r}) Y_{n,m}(\hat{\mathbf{s}})$$

- Performing $\int_{4\pi} d\Omega$ and $\int_{4\pi} \hat{\mathbf{s}} d\Omega$ on $I_d(\mathbf{r}, \hat{\mathbf{s}})$, we obtain

$$I_d(\mathbf{r}, \hat{\mathbf{s}}) \simeq \frac{U_d(\mathbf{r})}{4\pi} + \frac{3}{4\pi} \mathbf{J}_d(\mathbf{r}) \cdot \hat{\mathbf{s}}$$

Fick's law for the diffusion approximation

- Substituting the diffusion approximation in the RTE, and performing $\int_{4\pi} \hat{\mathbf{s}} d\Omega$, we obtain

$$\frac{\partial}{\partial t} J_d(\mathbf{r}, t) + (\mu_a + \mu'_s) J_d(\mathbf{r}, t) + \frac{1}{3} \nabla U_d(\mathbf{r}, t) = 0$$

where $\mu'_s = (1 - g) \mu_s$ and we define $\mu'_t = \mu'_s + \mu_a$
 $l'_t = 1/\mu'_t$

- If we assume $\frac{l'_t}{c} \left| \frac{\partial}{\partial t} J_d(\mathbf{r}, t) \right| \ll |J_d(\mathbf{r}, t)|$, we obtain

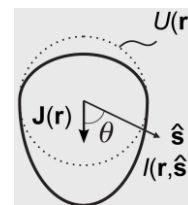
$$J_d(\mathbf{r}, t) = -D \nabla U_d(\mathbf{r}, t) \quad \text{where } D = \frac{1}{3(\mu_a + \mu'_s)}$$

The diffusion equation

$$\frac{\partial}{\partial t} U_d(\mathbf{r}, t) + \mu_a U_d(\mathbf{r}, t) - \nabla \cdot (D \nabla U_d(\mathbf{r}, t)) = S_0(\mathbf{r}, t)$$

If the diffusion coefficient $D = \frac{1}{3(\mu_a + \mu'_s)}$ is space-invariant, we obtain

$$\frac{\partial}{\partial t} U_d(\mathbf{r}, t) + \mu_a U_d(\mathbf{r}, t) - D \nabla^2 U_d(\mathbf{r}, t) = S_0(\mathbf{r}, t)$$



Time-domain Green's function

- The solution in an infinite, uniform medium for a delta function source $\delta(t - t')\delta(\mathbf{r} - \mathbf{r}')$

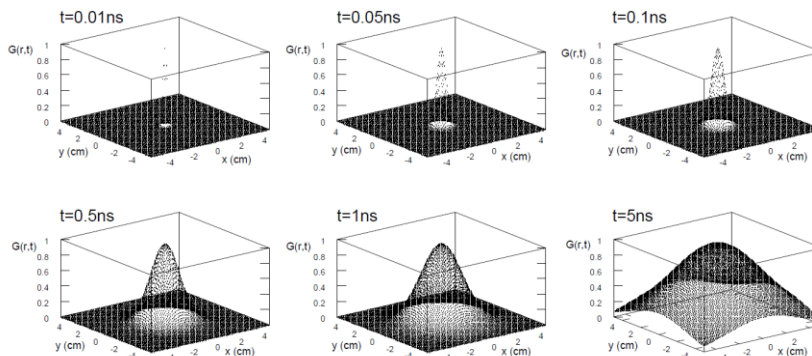
$$G(\mathbf{r} - \mathbf{r}', t - t') = c_0 \frac{\exp[-\mu_a c_0(t - t')]}{(4\pi D c_0(t - t'))^{3/2}} \exp\left(\frac{-|\mathbf{r} - \mathbf{r}'|^2}{4D c_0(t - t')}\right)$$

- *The solution is not causal
- For a general source term $f(\mathbf{r}, t)$, the solution is obtained by convolution

$$U(\mathbf{r}, t) = \int_0^\infty \int_V G(\mathbf{r}, \mathbf{r}', t, t') f(\mathbf{r}', t') d^3 r' dt'$$

Time-domain Green's function

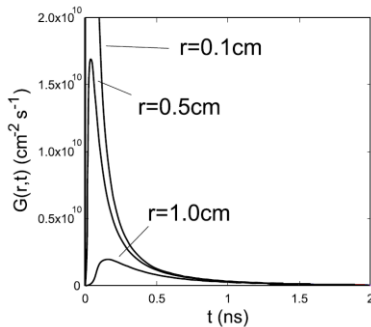
$$G(\mathbf{r} - \mathbf{r}', t - t') = c_0 \frac{\exp[-\mu_a c_0(t - t')]}{(4\pi D c_0(t - t'))^{3/2}} \exp\left(\frac{-|\mathbf{r} - \mathbf{r}'|^2}{4D c_0(t - t')}\right)$$



For $\mu_a = 0.25 \text{ cm}^{-1}$ $\mu'_s = 10 \text{ cm}^{-1}$ and $n=1.33$

Time-domain Green's function

$$G(\mathbf{r}-\mathbf{r}', t-t') = c_0 \frac{\exp[-\mu_a c_0(t-t')]}{(4\pi D c_0(t-t'))^{3/2}} \exp\left(\frac{-|\mathbf{r}-\mathbf{r}'|^2}{4D c_0(t-t')}\right)$$



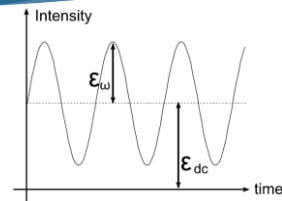
► Neglecting absorption, the time it takes the peak of the intensity to travel a distance r is $t = \frac{r^2}{6Dc_0}$

► If we wait long enough, the intensity will be spatially constant

For $\mu_a = 0.025 \text{ cm}^{-1}$ $\mu'_s = 10 \text{ cm}^{-1}$ and $n=1.33$

Frequency-domain Green's function

$$\epsilon(\mathbf{r}, \hat{\mathbf{s}}, t) = \epsilon_{dc}(\mathbf{r}, \hat{\mathbf{s}}) + \epsilon_\omega(\mathbf{r}, \hat{\mathbf{s}}) \exp(-i\omega t)$$



$$\frac{-i\omega}{c} U_\omega(\mathbf{r}) + \mu_a U_\omega(\mathbf{r}) - D \nabla^2 U_\omega(\mathbf{r}) = S_0(\mathbf{r}, \omega)$$

$$G_\omega(\mathbf{r}-\mathbf{r}') = \frac{\exp(i\kappa_\omega |\mathbf{r}-\mathbf{r}'|)}{4\pi D |\mathbf{r}-\mathbf{r}'|}$$

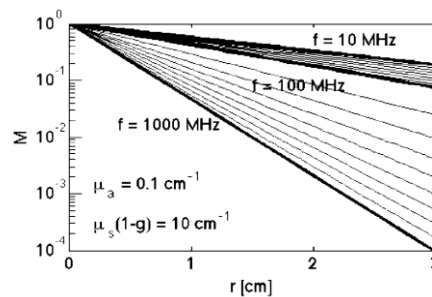
where

$$\kappa_\omega = \sqrt{-\frac{\mu_a}{D} + i \frac{\omega}{D c_0}}$$

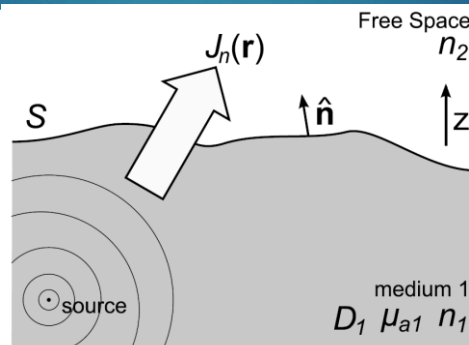
Effective attenuation coefficient

$$G_{\omega}(\mathbf{r} - \mathbf{r}') = \frac{\exp(i\kappa_{\omega}|\mathbf{r} - \mathbf{r}'|)}{4\pi D|\mathbf{r} - \mathbf{r}'|} \quad \kappa_{\omega} = \sqrt{-\frac{\mu_a}{D} + i\frac{\omega}{Dc_0}}$$

- For unmodulated light, the effective attenuation is given by $\mu_{eff} = \sqrt{\frac{\mu_a}{D}} = \sqrt{3\mu_a(\mu_a + \mu'_s)}$
- Modulation starts to play a role at $\omega = c_0\mu_a$. For $\mu_a = 0.1 \text{ cm}^{-1}$ and $n=1.33$, this means $f = 360 \text{ MHz}$
- At high frequencies, diffusion is responsible for attenuation (not absorption)



Boundary conditions for diffusive interface



- For index matched media, we require that the inward flux through $\hat{n}$ be zero:

$$\int_{\hat{\mathbf{s}} \cdot \hat{\mathbf{n}} < 0} I(\mathbf{r}, \hat{\mathbf{s}}) \hat{\mathbf{s}} \cdot \hat{\mathbf{n}} d\Omega = 0$$

Boundary conditions for diffusive interface

$$\int_{\hat{\mathbf{s}} \cdot \hat{\mathbf{n}} < 0} I(\mathbf{r}, \hat{\mathbf{s}}) \hat{\mathbf{s}} \cdot \hat{\mathbf{n}} d\Omega = \int_{\hat{\mathbf{s}} \cdot \hat{\mathbf{n}} < 0} \left[\frac{1}{4\pi} U_d(\mathbf{r}) + \frac{3}{4\pi} \mathbf{J}_d(\mathbf{r}) \cdot \hat{\mathbf{s}} \right] (\hat{\mathbf{s}} \cdot \hat{\mathbf{n}}) d\Omega$$

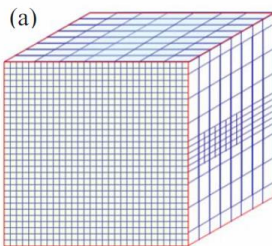


Robin boundary condition: $U_d(\mathbf{r}) - 2\mathbf{J}_d(\mathbf{r}) \cdot \hat{\mathbf{n}} = 0$

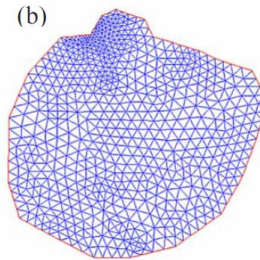
Generalization exists for non-index-matched media

Numerical solutions to the diffusion equation

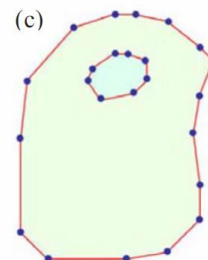
Finite
difference
method



Finite
element
method



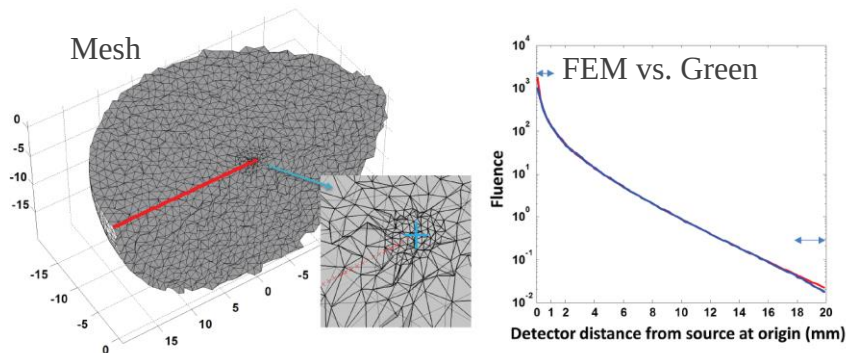
Boundary
element
method



S. L. Jacques and B. W. Pogue, JBO **13** (2008)

Finite difference method

The advantage of adaptive meshing: Better fit to boundary and source geometry



Pouyan Mohajerani, "Robust methods and algorithms for fluorescence imaging and tomography," Ph.D. Dissertation 2014

Online resources for FEM

web4.cs.ucl.ac.uk/research/vis/toast/start.html



TOAST++

Image Reconstruction in Diffuse Optical Tomography

Home News Download Documentation Support

sourceforge

Toastpp
Forward and inverse modelling in optical tomography
through to the source

Summary Files Reviews Support Wiki Code Tickets Discussion

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Description
Toast++ is a C++ software package for modelling light transport in scattering media using the finite element method. It allows for reconstruction of absorption and scattering distribution inside an object from boundary measurements of light transmission, using steady state, time domain or frequency domain information. Parallel and Python bindings exist for ease of use and fast prototyping.

Categories
sourceforge.net/projects/toastpp/files/latest/download?source=directory

The Toast++ repository on Sourceforge

The TOAST-Matlab help system

After installation of the Toast toolbox and setting up the required search paths with the mtoast_install script, a list of available toast functions can be obtained by typing

```
help toast
```

at the Matlab prompt. To get the Toast toolbox help page, type

```
doc toast
```

or open "Matlab Help" from the Matlab menu, and select the "TOAST Toolbox" entry. The help page contains links to function listings, demos, and introductory pages, as well as the Toast web home page. A collection of Toast demos is available by typing

```
demo toolbox toast
```

at the command prompt, by selecting "Demos" from the Matlab "Start" button, or from the "Demos" tab in the help browser. The demos are showing some of the Toast toolbox capabilities in small example problems.

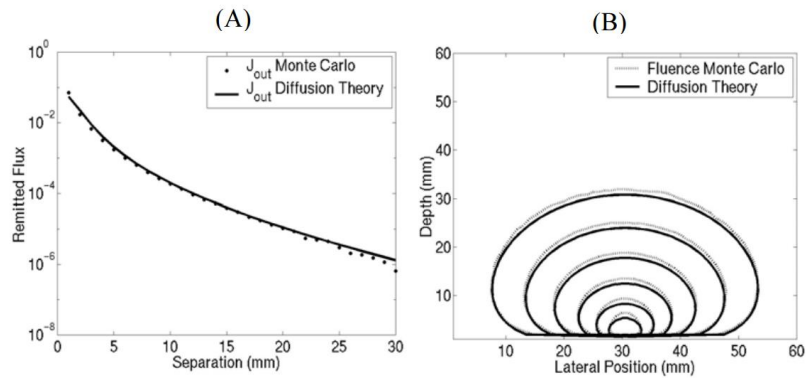
A first reconstruction test

To run a simple test reconstruction, follow these steps:

- Run matlab, and type

```
recon_001
```

Diffusion vs. Monte Carlo in CW



D.A. Boas *et al.* Opt. Express **10** (2002)

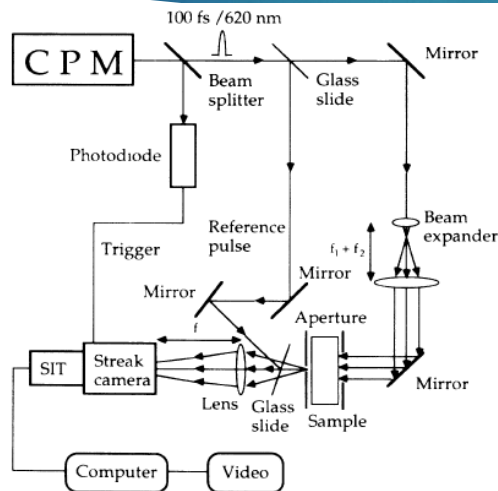
VOLUME 64, NUMBER 22

PHYSICAL REVIEW LETTERS

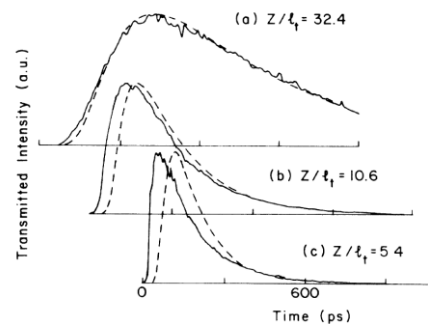
28 MAY 1990

When Does the Diffusion Approximation Fail to Describe Photon Transport in Random Media?

K. M. Yoo, Feng Liu, and R. R. Alfano

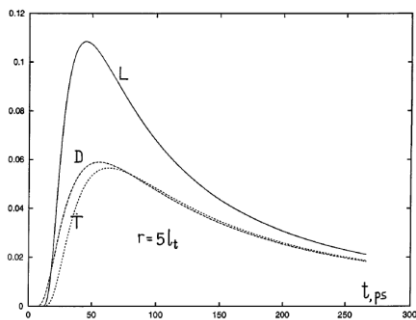


Experiment vs. the diffusion model



Longitudinal vs. transverse diffusion

$$I(\mathbf{r}, \mathbf{s}) = \delta(\mathbf{s} - \mathbf{s}_0) \delta(\mathbf{r})$$



A. Y. Polysuchuk and R. R. Alfano **21** (1996)

